

ROYAL FITNESS CLUB

RFC PREMIUM GUIDE #13

PRE-WORKOUT OPTIMISATION GUIDE

**Evidence-Based Performance Enhancement Before Every
Session**

5 Chapters · Science-Based · Custom Stack Builder · All Levels

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DISCLAIMER

This guide is for educational purposes only. Not medical advice. Consult a qualified healthcare professional before starting any protocol. Individual results vary.

CHAPTER 1 — PRE-WORKOUT SCIENCE

"A great pre-workout is not a product you buy — it is a system you build: food timing, caffeine, blood flow, and mental preparation."

The pre-workout window (60–90 minutes before training) represents the highest-leverage nutritional and supplementation opportunity in the day. Optimal pre-workout preparation maximises training performance, which in turn maximises the training stimulus, which drives muscle growth and fat loss. The most evidence-based pre-workout compounds are inexpensive, well-researched, and available individually — not bundled into expensive proprietary blends.

Compound	Evidence Grade	Effect	Dose	Timing
Caffeine	A (very strong)	Performance +3–12%, focus, fat oxidation	3–6mg/kg BW	45–60 min pre
Creatine	A (very strong)	Strength, power, muscle volume	5g daily	Any time (consistent)
Beta-Alanine	B (strong)	Muscular endurance, reduces fatigue	3.2–6.4g/day	Daily loading
Citrulline Malate	B (strong)	Blood flow, pump, endurance	6–8g	60 min pre
Beetroot / NO3	B (good)	Nitric oxide, endurance	500mg NO3 equiv	60–90 min pre
Tyrosine	C (moderate)	Mental focus, stress resistance	500mg–2g	60 min pre
Alpha-GPC	C (moderate)	Acetylcholine, mind-muscle	300–600mg	60 min pre

CHAPTER 2 — CAFFEINE OPTIMISATION

"Caffeine tolerance is a real phenomenon. Using it strategically — not habitually — is the difference between a training tool and a dependency."

Caffeine is the single most evidence-backed sports performance compound available. At appropriate doses, it increases strength by 3–7%, endurance by 5–12%, and fat oxidation by 10–15% during exercise. However, chronic daily use leads to complete tolerance within 1–2 weeks, eliminating virtually all performance benefit. Strategic use — reserved for training days with mandatory caffeine-free days — preserves the ergogenic effect.

User Type	Daily Caffeine	Recommended Strategy	Pre-Workout Dose
Non-user / beginner	< 50mg	Start at 100mg pre-workout	100–150mg
Moderate user	100–200mg (1–2 coffees)	No coffee on training days	150–200mg pre
Heavy user (> 3 coffees)	300–500mg	Caffeine cycling protocol	200–300mg pre (after 5-day break)
Elite protocol	Minimal daily use	5 days off caffeine, then 3–6mg/kg pre-competition	3–6mg/kg body weight

■ **Caffeine cycling protocol: 5 caffeine-free days every 4 weeks restores full sensitivity. Use decaf coffee, green tea (low caffeine) or nothing on off days.**

CHAPTER 3 — NITRIC OXIDE & BLOOD FLOW

Nitric oxide (NO) is a vasodilator that increases blood flow to working muscles, resulting in improved nutrient and oxygen delivery, greater "the pump" sensation, and potentially enhanced hypertrophy through cell swelling mechanisms. L-Citrulline is the primary evidence-backed NO precursor.

Compound	Mechanism	Dose	Time to Effect	Synergies
L-Citrulline	Arginine precursor → NO via NOS	6g	60 min	Malate 2:1 ratio helps
Citrulline Malate	Same + malate reduces lactate	8g (6g citrulline)	60 min	Caffeine, beetroot
Beetroot Powder	Direct NO3 → NO2 → NO	500mg NO3	60–90 min	Citrulline synergy
L-Arginine	Direct NO precursor (poor bioavailability)	3–6g	45 min	Less effective than citrulline
Pine Bark Extract (Pycnogenol)	eNOS activation + antioxidant	200mg	30–45 min	Citrulline stack

CHAPTER 4 — NOOTROPICS FOR TRAINING

Mental focus, the mind-muscle connection, and psychological arousal are trainable variables that substantially impact training quality. The following nootropic compounds improve training performance through CNS mechanisms.

Nootropic	Effect	Dose	Safety Profile	Note
L-Tyrosine	Dopamine, noradrenaline precursor	500mg–2g	Excellent	Best under stress or sleep deprivation
Alpha-GPC	Acetylcholine precursor, power output	300–600mg	Good	
Huperzine A	AChE inhibitor — prolongs acetylcholine	50–100mcg	Moderate	Cycle 2 weeks on/2 off
Rhodiola Rosea	Adaptogen — reduces perceived effort	200–600mg	Excellent	Consistent use > single dose
DMAE	Precursor to choline	300–500mg	Good	May cause vivid dreams

CHAPTER 5 — BUILDING YOUR CUSTOM STACK

Custom Pre-Workout Tiers

Tier	Budget	Stack	Expected Benefit
Essential (low budget)	<■1000/month	Caffeine (200mg) + Creatine 5g	Performance +8%, strength +5%
Standard	■1000–3000/month	+ Citrulline 6g + Beta-Alanine 3.2g	Performance +15%, endurance +12%
Advanced	■3000–6000/month	+ Alpha-GPC + Beetroot + Tyrosine	Maximum focus + blood flow + performance
Elite	>■6000/month	Full stack + systematic caffeine cycling	Peak optimisation protocol

30-Minute Pre-Workout Protocol

- T – 90 min: Meal if using food-based pre-workout. 40g carbs + 30g protein
- T – 60 min: Citrulline 6g + Beta-Alanine 3.2g + Beetroot 500mg
- T – 45 min: Caffeine dose (capsule or black coffee)
- T – 20 min: Alpha-GPC or Tyrosine (cognitive focus)
- T – 5 min: Warm-up begins. Mental priming — set your session intention

■ *The best pre-workout ever created is not available in any store: 8 hours of quality sleep, a well-timed protein and carb meal, strong intention, and a playlist that elevates your state.*